Using ClamAV to Detect and Prevent Malware

Malware is computer software that can lead to serious disasters, ranging from vital data loss to a network security breach. It can infect a client computer or a server by means of data transfer, and only sophisticated antivirus software can scan and detect this in real-time. This article describes various types of malware that can affect computer systems, and how ClamAV can be used to mitigate their effects.

There is a lot of antivirus software available in the market for use, but not all of it is free or open source. One such open source software is Clam AntiVirus (ClamAV), which stands out because of its features and effectiveness. ClamAV was first developed for the UNIX operating system but now is also available for AIX, BSD, HP-UX, Linux, MacOS, OpenVMS, OSF (Tru64) and Solaris.

ClamAV does not include any graphical user interface (GUI); it uses only a command line interface. However, various third party developers have developed a GUI for this antivirus for different platforms. For Linux, the ClamTk GUI is used to work with the ClamAV command line antivirus and it's also free to use. ClamXAV is another GUI used for MacOS but it's not free (trialware).

For Windows, various GUI apps have been created, which include Immunet, ClamWin and Clam Sentinel. Immunet is a free cloud-based app that works with ClamAV, and is developed and maintained by Cisco. ClamWin is another free front-end made for Microsoft Windows by ClamWin Pty Ltd. This Windows GUI can make ClamAV do on-demand (user started) scanning, automatic updates, scan scheduling, context menu integration to Explorer, and an add-in for Microsoft Outlook. However, ClamWin does not provide on-

access scanning and needs additional software to do that. ClamAV can also be used with Mozilla Firefox using a plugin that works with ClamWin to scan downloaded files. Clam Sentinel is also a free GUI app for Microsoft Windows that works with ClamAV. It runs in the system tray and can detect any file system changes. It scans the modified files using ClamWin in real-time. It can run on Windows 98/98SE/ME/XP/Vista/7/8 and has a real-time scanner for ClamWin. It also features optional system change messages and proactive heuristic protection. Heuristic analysis is used by many antivirus software to detect unknown viruses using various decision rules and weighing methods. ClamAV, too, uses this type of protection.

ClamAV is mostly used for email scanning and is actually a framework, which provides a protection engine that can be connected to a Message Transfer Agent (MTA) like Sendmail, PostFix, etc. It's an excellent email gateway AV deployed by ISPs, telecom service providers, and Web hosting companies. It's also used by many AV gateway vendors like Barracuda and OS vendors like Apple.

Types of malware

Malware is a computer program or software that contains a bug (created intentionally or because of bad coding), which can cause harm or access to a computer system or data, without the consent of the user. Various types of malware include computer viruses, worms, Trojan horses, ransomware, spyware, adware, rogue software, wiper and scareware. Among these malware, computer viruses and Trojan horses are most commonly used to harm a computer, and are spread through Internet files and computer games.

- **Computer virus:** A virus is malware that can infect a computer file system from any infected file downloaded or saved in that computer. Just like a real biological virus, a computer virus can reside inside a file like an infection. The virus gets activated when that infected file is opened, and then it either changes or damages the data and infects other files in the computer. A virus requires a host file to which it writes its code. That's why most of the computer viruses can enter a computer though unreliable sources like pirated software and games, email attachments, and file transfers